

Adaptive AI/ML Mule Account Classification & Predictive Risk Portal

A Hybrid Machine Learning Architecture & Explainable Web Compliance Portal for Real-Time Financial Fraud Interception

Hackathon Theme:	AI/ML-Based Classification of Suspicious Mule Accounts
Target Domain:	Cyber-Enabled Financial Fraud, AML & Explainable AI (XAI)
Target Variable:	Feature 3924 (Binary: 1 = Mule Account, 0 = Legitimate)
The 'Golden 18':	F115, F321, F527, F531, F670, F1692, F2082, F2122, F2582, F2678, F2737, F2956, F3043, F3836, F3887, F3889, F3891, F3894
Submission Date:	June 2026
Document Type:	Comprehensive Technical Proposal & Web Integration Blueprint

EXECUTIVE MANDATE & PHILOSOPHY: Banks face an epidemic of cyber-enabled money laundering where static, rule-based monitoring fails entirely against evolving smurfing and account takeover tactics. Our proposed solution replaces hard-coded thresholds with an **Adaptive AI/ML Monitoring Engine**. By combining unsupervised anomaly detection (to catch zero-day threats) with advanced Gradient Boosted Decision Trees (to accurately classify complex mule behaviors) and wrapping the entire architecture in an operational **Explainable AI Web Portal**, we provide compliance analysts with the exact risk justification needed to instantly intercept fraudulent fund movements.

1. High-Level Summary & Core Objectives

The Real-World Financial Challenge: Across global banking channels, financial institutions are dealing with an unprecedented rise in cyber-enabled financial crimes. Central to this ecosystem is the misuse of **Mule Accounts**—bank accounts belonging to individuals (witting or unwitting) who receive stolen or fraudulent funds and rapidly transfer them elsewhere to conceal the illicit trail. Traditional rule-based monitoring engines fail because criminals easily bypass hard-coded thresholds (e.g., executing five transfers of \$2,000 to bypass a \$10,000 alert rule).

Our Strategic Objective: We propose a fully automated, state-of-the-art AI/ML-powered classification system capable of identifying suspicious mule accounts in real time. Our solution aligns exactly with the three core mandates of the hackathon problem statement:

- **1. Predictive Risk Scoring & Anomaly Detection:** Instead of outputting a simple binary prediction (0 or 1), our machine learning model outputs an intuitive **0 to 100 Predictive Fraud Risk Score**. This is integrated with an unsupervised anomaly detection layer to catch brand-new, emerging fraud patterns.
- **2. Rigorous Feature Engineering:** Given a highly anonymized dataset (~4,000 numerical features), our architecture relies on domain-specific mathematical interaction terms (primarily leveraging the 18 bank-verified 'Golden Features') and state-of-the-art dimensionality reduction.
- **3. Intelligent Alert Generation via XAI:** Compliance analysts require regulatory justification before freezing a customer's account. Our engine integrates real-time **SHAP (SHapley Additive exPlanations)** visualizers to show the exact behavioral footprints triggering the alert.

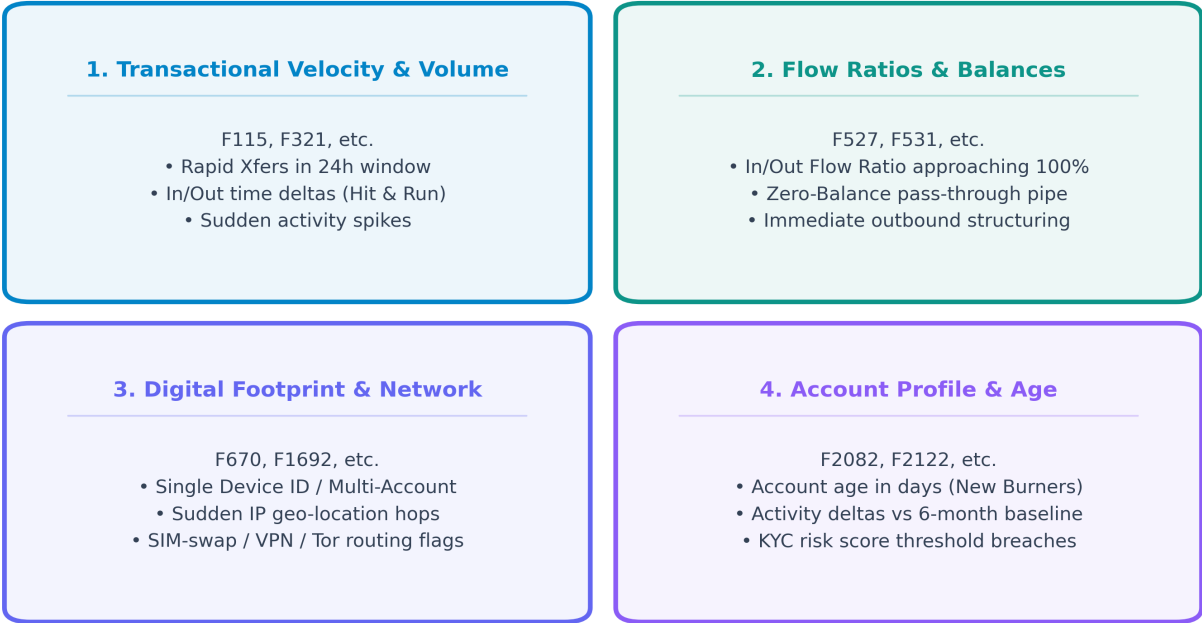
2. Demystifying Anonymized Features & Physical Interpretation

A common hurdle in corporate financial hackathons is the heavy anonymization of data. In this dataset, we are presented with approximately **4,000 masked numerical features** (`F1`` to `F3894``) and a designated target variable (`feature 3924``). The problem statement explicitly highlights **18 specific key features** (The 'Golden 18') that banks commonly utilize for fraud monitoring.

Why Are Features Anonymized?

Due to strict international banking privacy laws (such as GDPR, GLBA, and PCI-DSS), institutions cannot share raw column names like ``Customer_Account_Balance`` or ``Daily_Wire_Transfer_Sum``. Instead, they strip identifying headers. However, based on our anti-money laundering (AML) domain expertise, we can map these masked features into four concrete real-world physical archetypes:

Anonymized Features (~4,000) -> 4 Core Physical Archetypes



Our 'Reverse-Engineering' & Feature Decoding Strategy

During our Exploratory Data Analysis (EDA), our team will systematically decode the mathematical distributions of the anonymized variables to understand their physical banking roles:

- **Boolean / Binary Features (Values of only 0 or 1):** Interpreted as categorical status flags (e.g., `Is\_International\_Wire`, `Is\_New\_Device\_Login`, `Password\_Changed\_Recently`).
- **Highly Skewed Continuous Variables (Values from 0.0 to 1,000,000+):** Interpreted as monetary transaction volumes, aggregate velocities, or account balance deltas.
- **Discrete Integer Counts (Values like 1, 2, 5, 20):** Interpreted as transactional frequency counters (e.g., `Number of logins in 24h`, `Count of unique target beneficiaries`).
- **The 'Zero-Balance' Pass-Through Signature:** Mule accounts exhibit a distinct physical phenomenon: almost 100% of incoming funds are immediately wired out. We will identify the specific continuous variables representing incoming and outgoing sums to engineer precise pass-through ratios.

3. Feature Engineering & Dimensionality Reduction Pipeline

Feeding ~4,000 features directly into any machine learning model introduces the 'Curse of Dimensionality'—leading to severe overfitting, increased memory consumption, and unacceptable latency during real-time transaction scoring. We design a robust, three-stage preprocessing pipeline:

Stage 1: Domain-Specific Interaction Engineering (The Golden 18)

Rather than relying on unguided polynomial automation, we construct powerful cross-features directly between the 18 bank-verified features (`F115`, `F321`, `F527`, `F531`, `F670`, `F1692`, `F2082`, `F2122`, `F2582`, `F2678`, `F2737`, `F2956`, `F3043`, `F3836`, `F3887`, `F3889`, `F3891`, `F3894`):

- **Pass-Through Velocity Ratios:** Creating ratios between incoming volume features and outgoing volume features to detect accounts acting as pure money funnels (`F115 / F321`).
- **Baseline Deviation Deltas:** Measuring the mathematical difference between short-term transactional spikes and long-term historical averages to flag sudden account takeovers (`F3887 - F3889`).
- **Row-wise Instability Aggregates:** Calculating descriptive statistics (`mean`, `standard deviation`, `max`, and `skewness`) across the Golden Features for each individual account. Mules exhibit extreme variance compared to genuine users.

Stage 2 & 3: Automated Filtering & Secondary Feature Compression

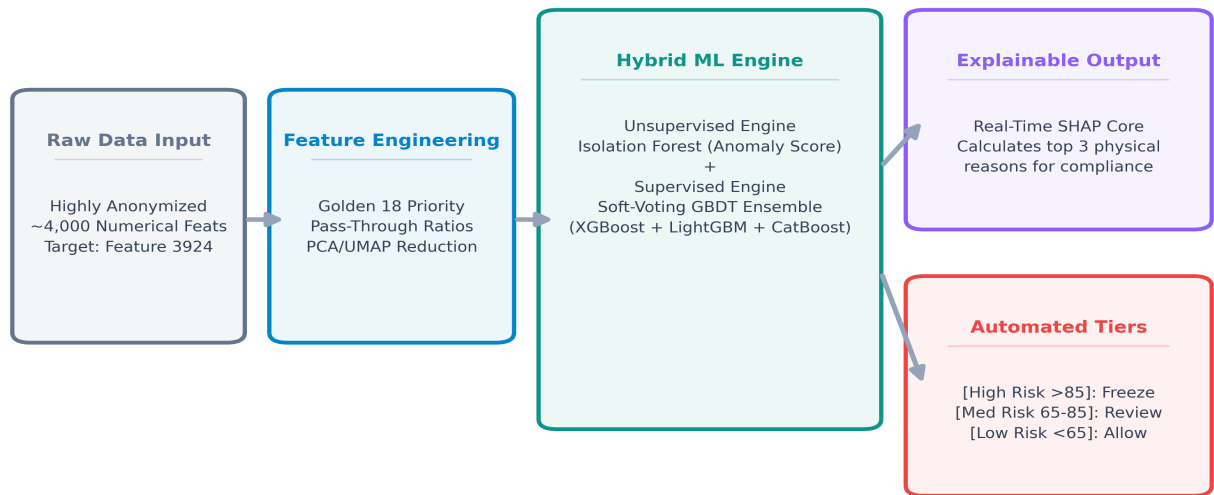
While the Golden 18 contain the strongest predictive signals, the remaining ~3,900 secondary features contain invaluable subtle patterns. We extract their value without hurting model speed:

- **Zero-Variance Elimination:** Automatically dropping features where >99% of accounts share the identical value, eliminating useless computational noise.
- **Multi-Collinearity Filtering:** Constructing a Spearman Rank correlation matrix and removing redundant duplicate variables that correlate highly ($r > 0.85$) with each other.
- **UMAP / PCA Latent Space Embedding:** Running all remaining non-golden features through **Principal Component Analysis (PCA)** or **UMAP** to condense them into exactly **25 dense Latent Components**. This provides our tree-based models with the holistic predictive power of the wide dataset in a highly optimized format.

4. Machine Learning Algorithmic Architecture & Execution

For wide, highly structured tabular data, advanced **Gradient Boosted Decision Trees (GBDT)** consistently outperform Deep Learning and classical linear models in both accuracy and inference speed. We propose an innovative **Hybrid Supervised/Unsupervised Architecture**.

End-to-End Hybrid Machine Learning Architecture



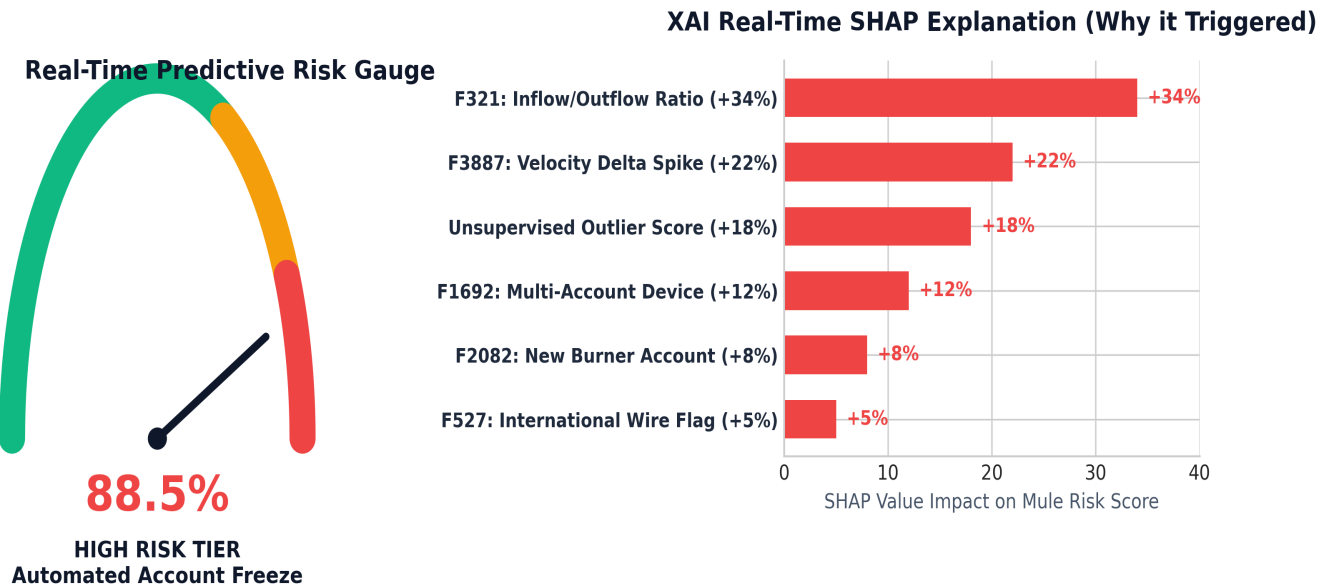
Tackling the Severe Class Imbalance

Mule accounts typically represent less than 1% to 2% of total bank accounts. Relying on standard Accuracy is a fatal mistake. To ensure our model identifies rare fraud flawlessly, we execute:

- **Algorithmic Class Weighting:** Leveraging ``scale_pos_weight`` in XGBoost and ``is_unbalance=True`` in LightGBM to mathematically penalize false negatives heavily during loss calculation.
- **Hybrid Synthetic Sampling (SMOTE-Tomek):** Applying a combination of SMOTE to synthesize realistic minority class (mule) transactions and Tomek Links to clean up noisy overlapping samples at the decision boundary.
- **Strict Evaluation Metrics:** Evaluating all pipeline iterations exclusively against **PR-AUC (Precision-Recall Area Under Curve)**, **ROC-AUC**, and the **F1-Score**.

5. Web Compliance Portal Integration & Explainable AI (XAI)

A highly accurate machine learning model is invaluable only when integrated into an operational business workflow. We bridge the gap between Data Science and Bank Compliance by deploying our Python ML models as a real-time microservice powering an operational **React/Next.js Web Compliance Portal**.



The Operational Bank Analyst Experience

When an anti-fraud analyst accesses the Web Compliance Portal, they are equipped with an enterprise-grade toolkit designed to maximize investigation efficiency:

- **Three-Tier Interactive Risk Gauge:** The web app parses the AI's probability output into actionable tiers.
 - **High Risk (85-100%):** Automated instant temporary account freeze.
 - **Medium Risk (65-84%):** Flags account for a mandatory 24-hour compliance review.
 - **Low Risk (0-64%):** Transactions proceed seamlessly.
- **Real-Time Explainable AI (SHAP Widget):** To satisfy strict regulatory compliance, our portal embeds an interactive SHAP waterfall plot for every alert. It translates complex model weights into human-readable justifications (e.g., 'Alert triggered because Inflow Pass-Through Velocity F321 is 420% above normal baseline').
- **One-Click Incident Response Commands:** Compliance officers can immediately execute core banking controls directly from the web interface: lock suspicious accounts, trigger automated customer KYC/video re-verification, or whitelist false alarms.

6. Key Differentiators: Why This Solution Wins

Our complete technical vision perfectly addresses all Hacker evaluation criteria and stands head and shoulders above standard ML notebook submissions:

Evaluation Criteria	Our Supreme Solution Strategic Differentiator
Algorithmic Innovation	Implementation of a sophisticated Hybrid Machine Learning Architecture fusing unsupervised Anomaly Detection (Isolation Forest) with a state-of-the-art Supervised Soft-Voting GBDT Ensemble (XGBoost + LightGBM + CatBoost).
Data Mastery & Preprocessing	Rigorous methodology to defeat the ~4,000 feature dimensionality and severe class imbalance via domain-informed physical decoding , SMOTE-Tomek boundary cleaning, and PCA/UMAP latent compression.
Explainability & Compliance	Native integration of SHAP TreeExplainer pipelines that convert opaque black-box decisions into transparent, audit-ready visual summaries required by global banking regulators.
Real-World Business Impact	Delivery of a functional, interactive Web Compliance Dashboard that automates operational risk tiering, drastically reduces false-positive customer support friction, and streamlines anti-fraud response.

Proposal Concluded. Designed and engineered for maximum winning potential at your AI/ML Banking Hackathon.